

DETERMINANTS AND PREDICTIVE MODELING OF H1N1 AND SEASONAL INFLUENZA VACCINATION UPTAKE

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DATA SCIENCE

PT 14

PHASE THREE PROJECT

OVERVIEW

Vaccination rates for H1N1 and seasonal influenza remain inconsistent across populations. This project analyzes factors influencing H1N1 and seasonal influenza vaccination uptake using survey data and also Improves outbreak prevention as an interventions before diseases like H1N1 and seasonal flu spread widely.

BUSINESS AND DATA UNDERSTANDING

The dataset was obtained from DrivenData with 26,707 records and 36 features. Analyses was based on vaccination distribution, education, perceived effectiveness and risk perceptions as factors influencing vaccine uptake

BUSINESS AND DATA UNDERSTANDING Cont...

Data preprocessing involved:-

- Handling missing values,
- Encoding categorical variables,
- Scaling numerical features.
- The dataset was split into training and validation sets (80/20),

Pipeline was used to ensure consistent preprocessing and prevent data leakage.

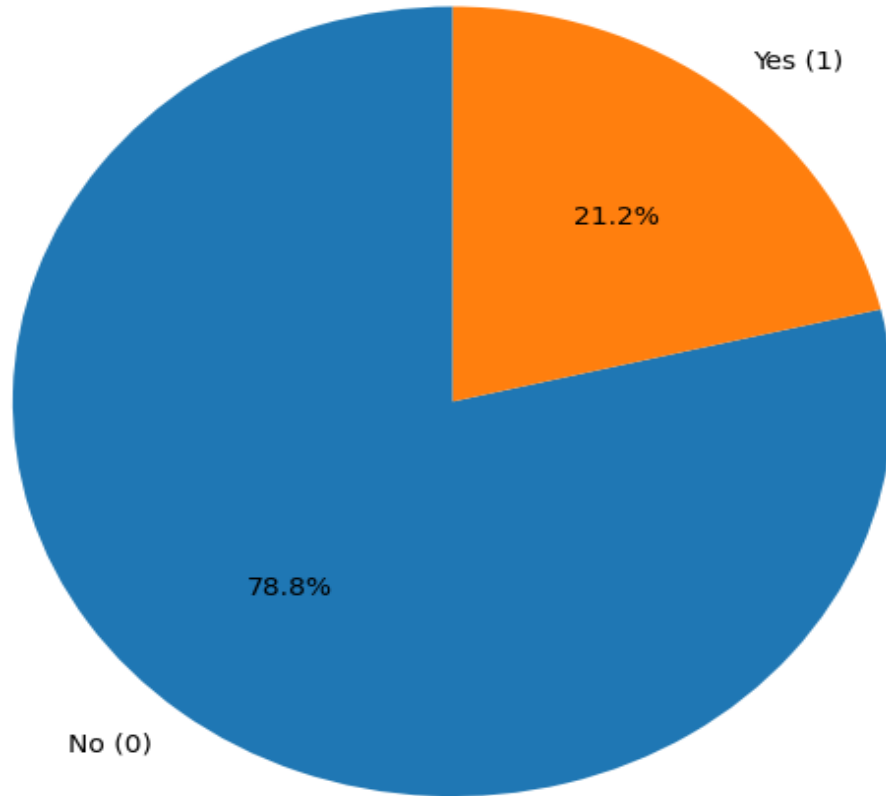
STAKEHOLDERS

- Ministry of Health
- Healthcare providers
- NGOs
- General public

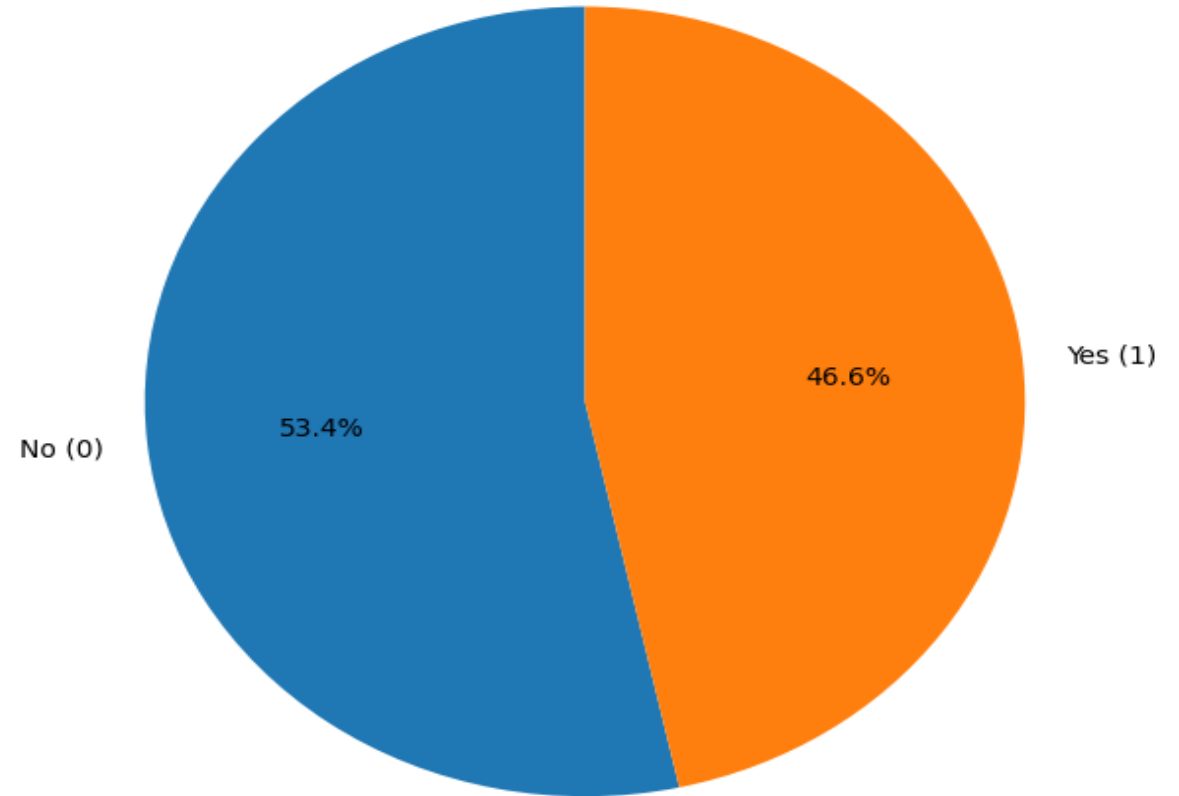
Visualization

Distribution of H1N1 and seasonal influenza vaccination uptake

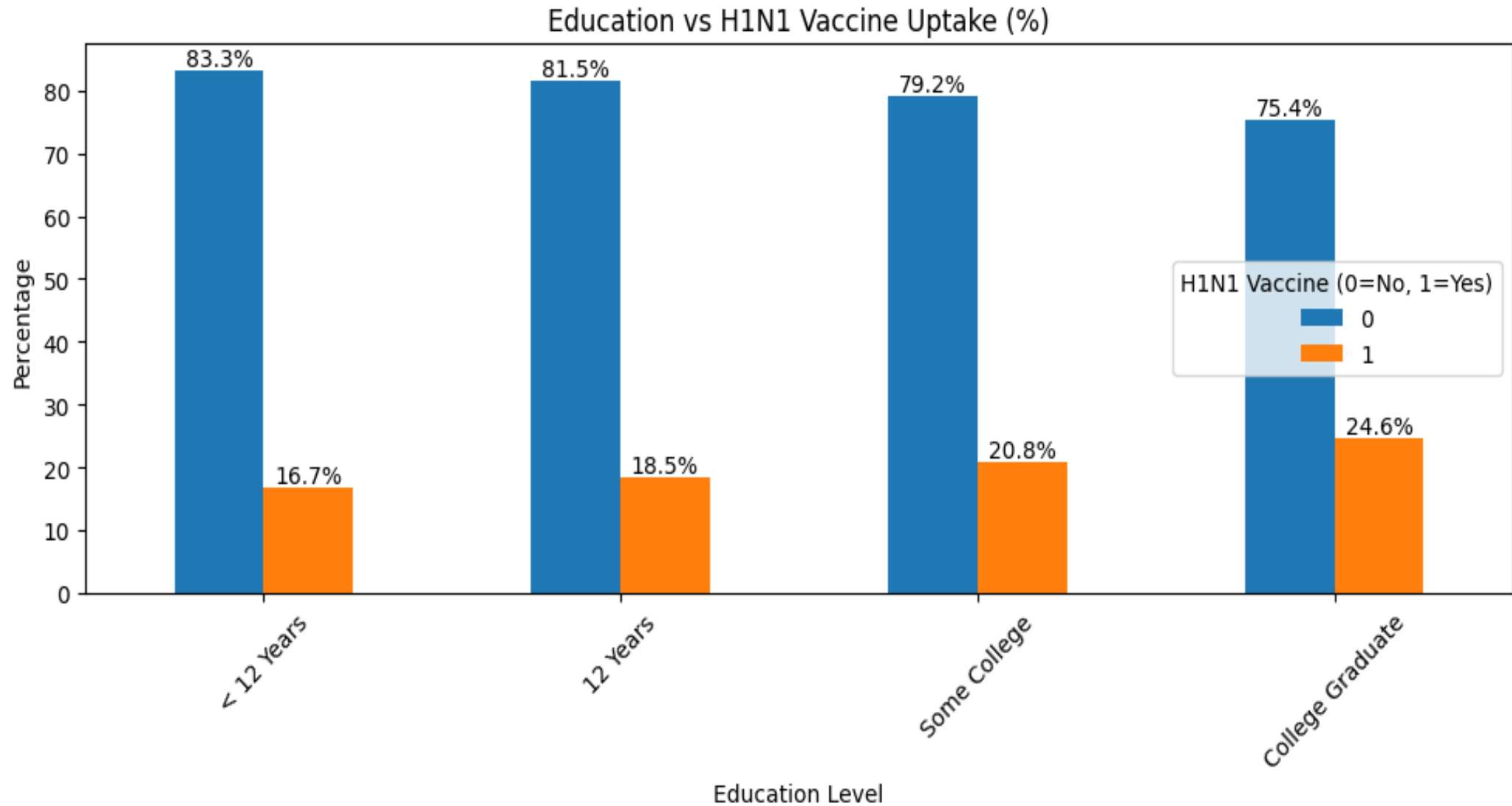
H1N1 Vaccine Distribution



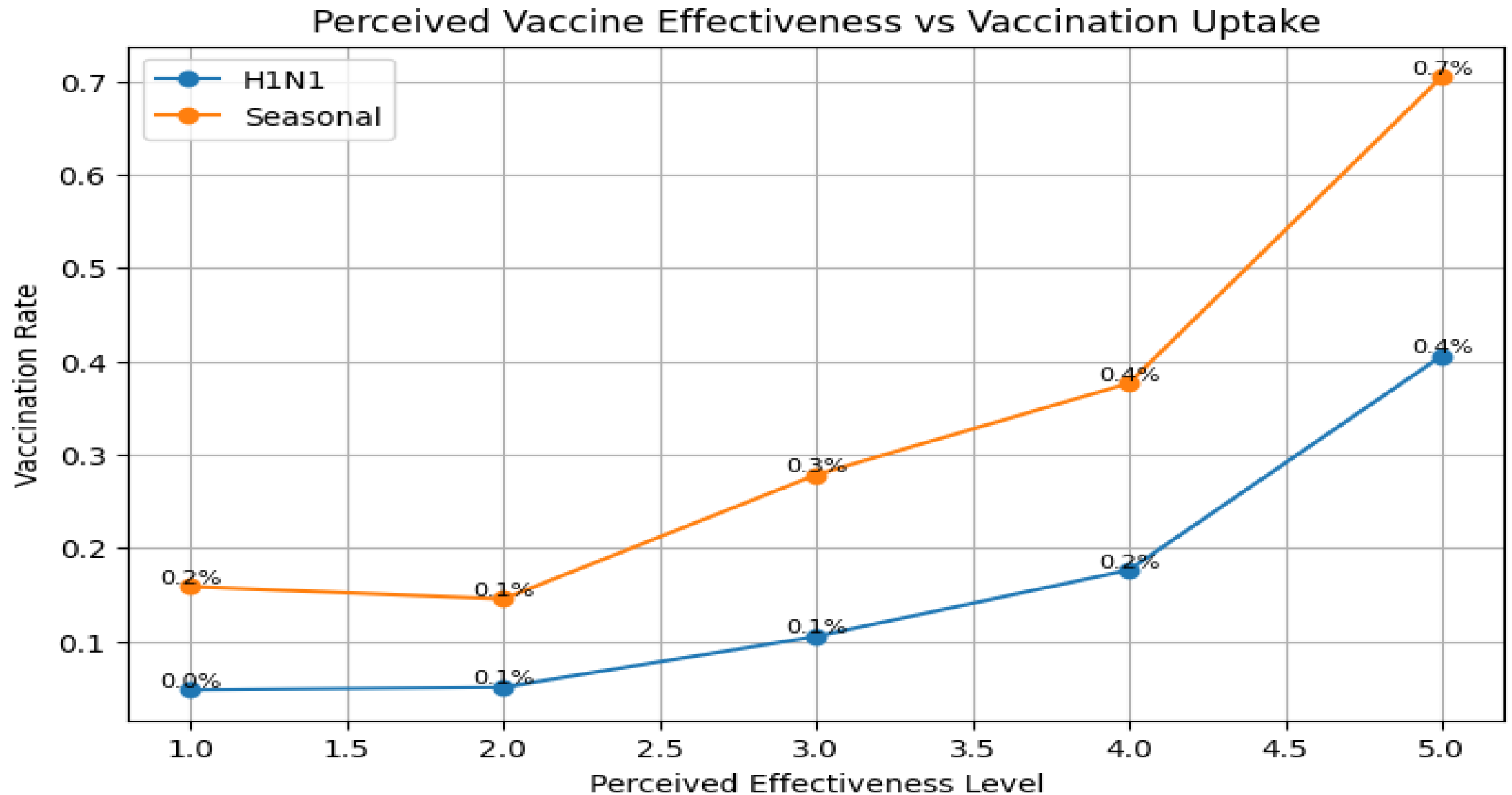
Seasonal Flu Vaccine Distribution



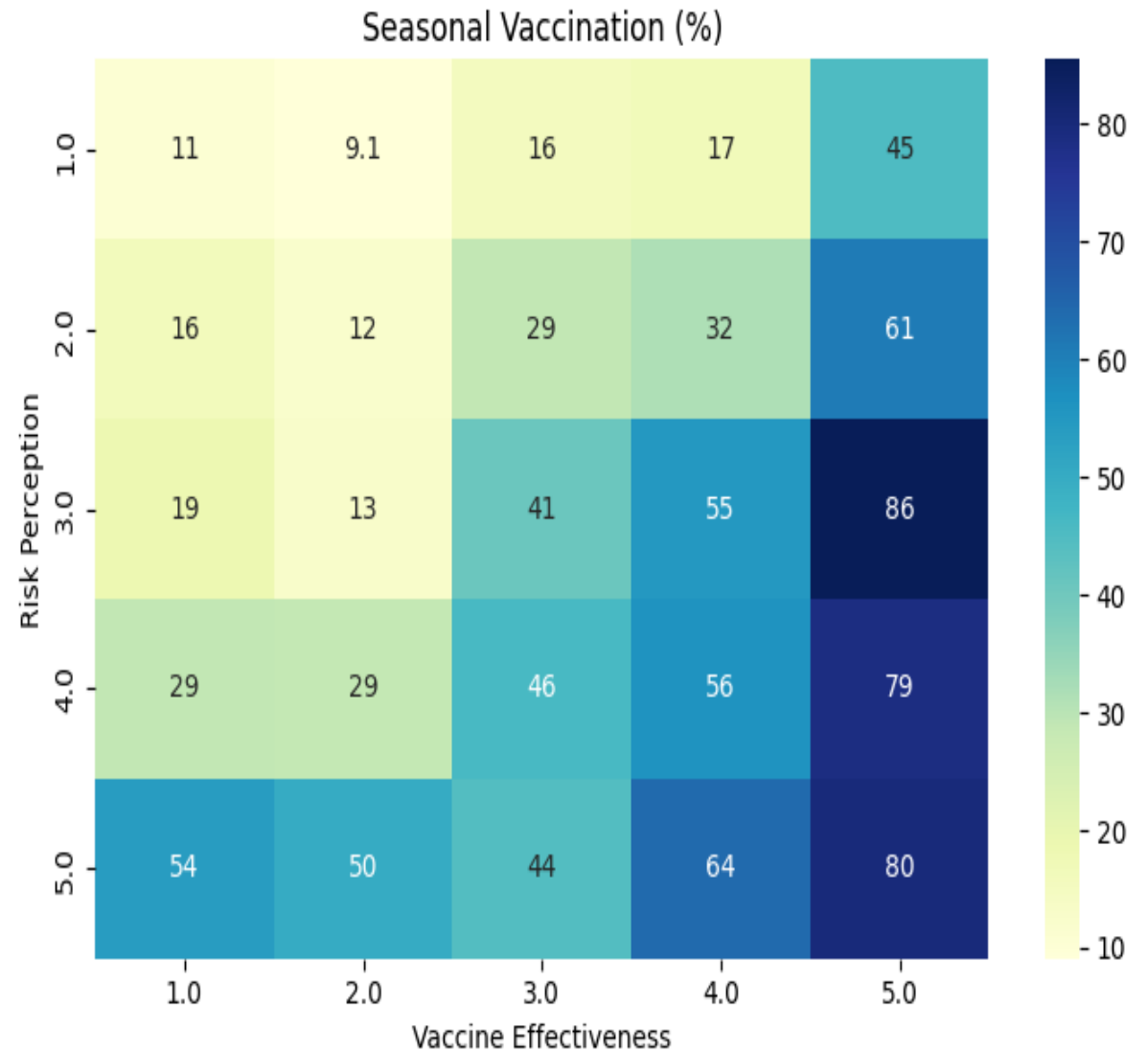
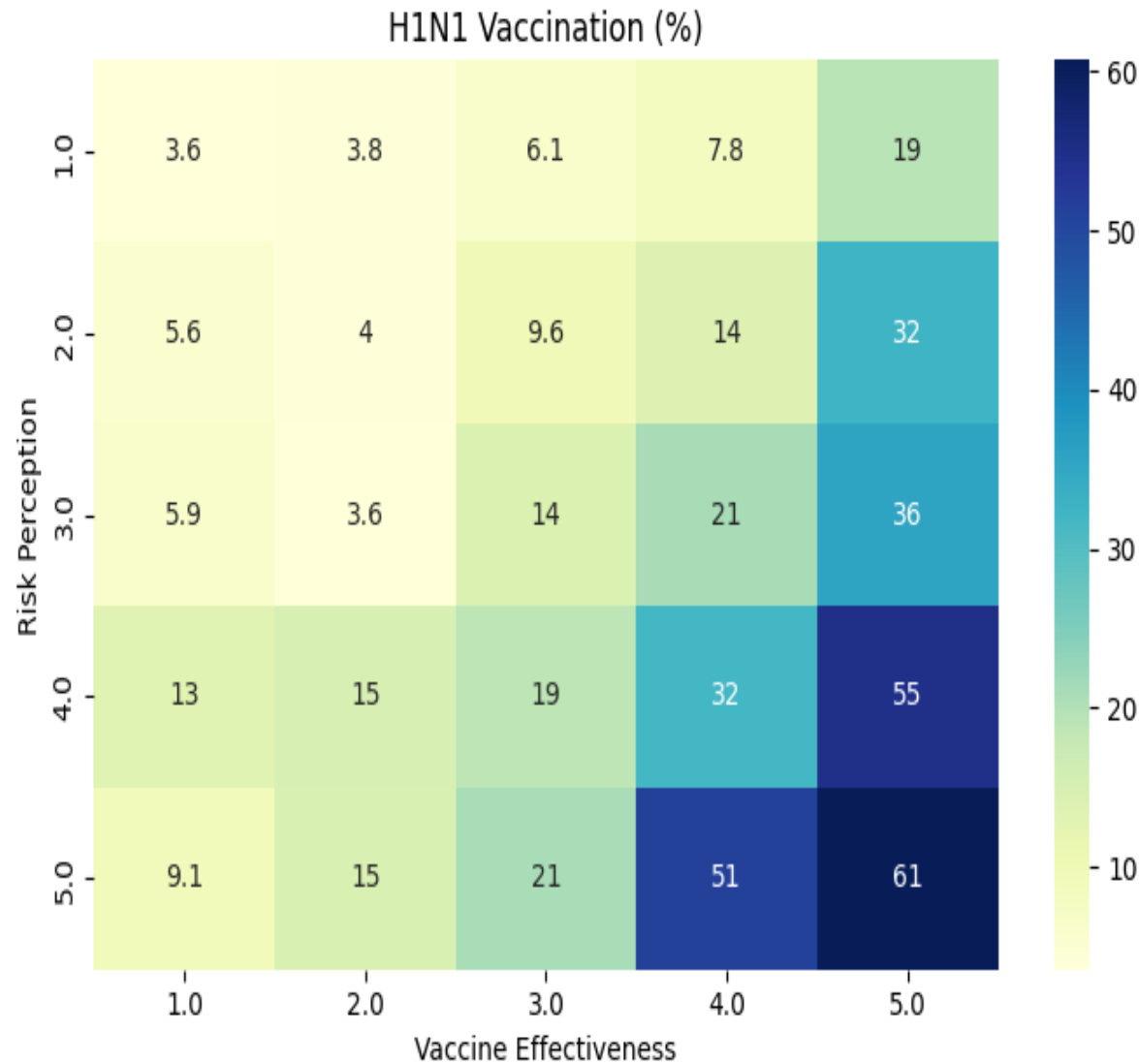
Relationship between education factors and vaccine uptake



Effect of perceived vaccine effectiveness on vaccination uptake



Effect of perceived disease risk and vaccine effectiveness on H1N1 and seasonal influenza vaccination uptake



INSIGHT

- Seasonal flu vaccination is more routine while H1N1 may be perceived as less familiar
- Higher education levels show increased vaccination uptake, suggesting that awareness and knowledge influence vaccine acceptance

INSIGHT Cont ...

- Low perceived effectiveness is a major barrier to vaccination uptake. People who believe vaccines are more effective are significantly more likely to get vaccinated.
- Across both H1N1 and seasonal flu, as risk perception and vaccine effectiveness vaccination rates increase.

MODELLING

Machine Learning Models Used:-

Random Forest this captures complex relationships in the data, handles non-linearity and feature interactions effectively

Logistic Regression is highly interpretable and easy to explain

Decision Tree is simple and highly interpretable model

EVALUATION

Random Forest

- Highest accuracy (H1N1: 0.82, Seasonal: 0.78)
- Best for overall predictions
- Lower recall → may miss some positive cases

Logistic Regression

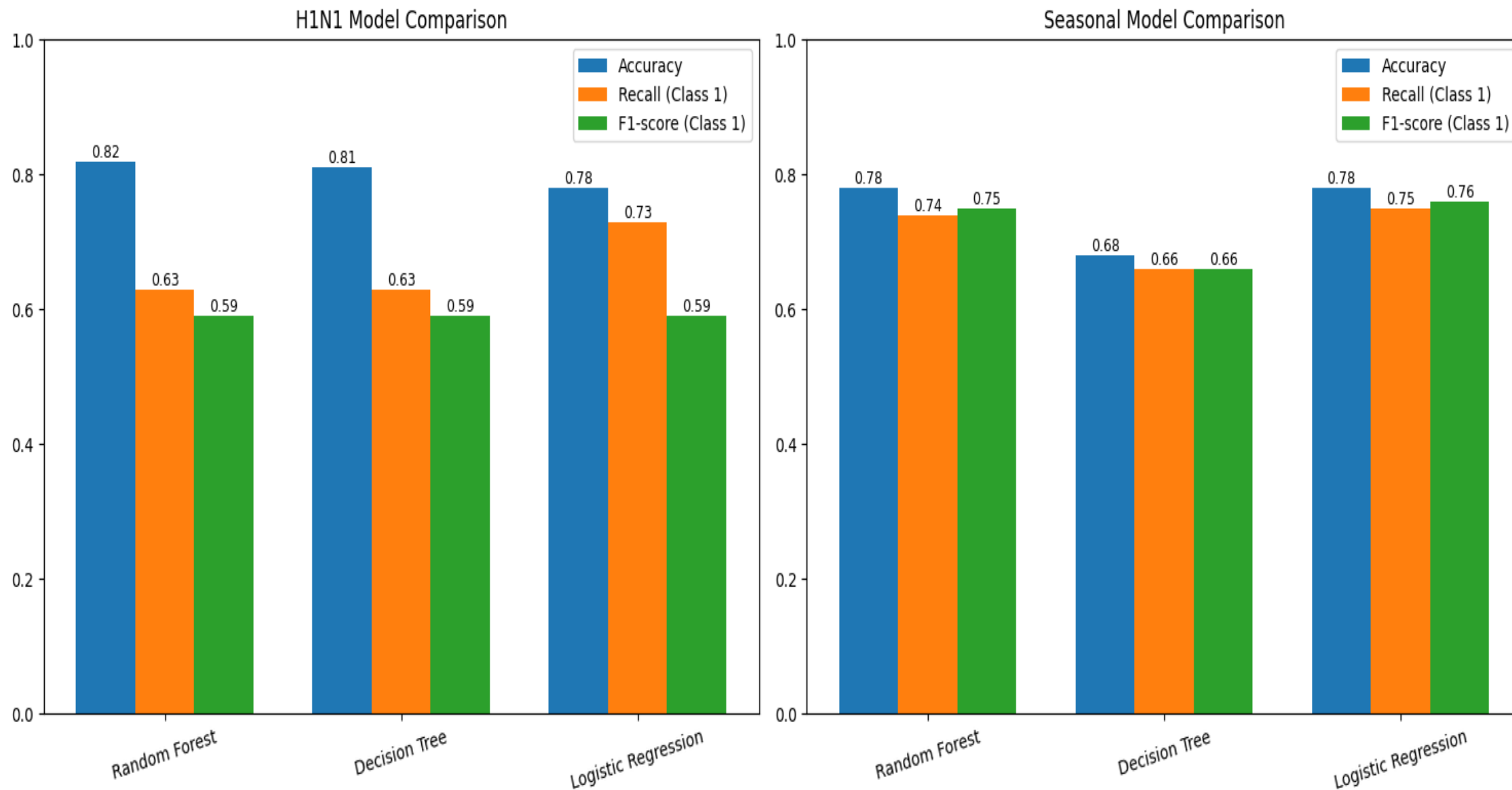
- ❖ Highest recall (H1N1: 0.73, Seasonal: 0.75)
- ❖ Most balanced performance (Seasonal F1-score: 0.76)
- ❖ More reliable for identifying positive cases

EVALUATION Cont..

Decision Tree

- Lowest performance overall
- Struggles especially on Seasonal Flu (accuracy: 0.68)
- Useful mainly as a baseline model

COMPARISON OF MODEL PERFORMANCE



LIMITATIONS

These limitations may reduce prediction accuracy and the effectiveness of model

- Data Quality Issues -High missing data this introduce bias and reduce reliability
- Class Imbalance between vaccinated vs non-vaccinated limits detection of positive cases
- Misclassification may affect targeting of vaccination efforts may be form risk of overlooking vulnerable populations

RECOMMEDATION

- Focus intervention on groups predicted as non-vaccinated and emphasize personal risk exposure for example age, occupation
- Awareness campaigns on low education population use of local languages, Use visual communication Community outreach programs for example use of charts

NEXT STEP

- Deploy targeted vaccination campaigns for high-risk groups
- Integrate model into public health systems to identify hesitant individuals
- Use clinics and community outreach for awareness programs
- Hyperparameter tuning for all models
- Apply resampling techniques for imbalance

THANK YOU